Penalties Sized to the Gain

When do random audits stop AI agents from over-using a shared resource?

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PhD catch-up
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The Project in One Slide

01 WHY

Oversight of AI agents is sampled

AI agents are starting to act on resources they share with others, such as compute, energy and budgets. Whoever oversees them can usually check only a few of their actions.

02 WHAT I BUILT

A shared resource with checks

Six agents draw on one renewable resource over many rounds. A reviewer sets how much each may take, and a few agents are checked at random each round.

03 WHAT I TESTED

Consequences and penalty size

What a caught agent should face (a fine, a memory of the offence, or both), and whether the fine that stops an agent can be predicted before it is tried.

04 WHAT I FOUND

Checks work through consequences

Checks reduced harm only when a catch had a consequence. The stopping fine could be predicted from how much an agent gains by over-taking, for programmed agents and for language models.

The rest of the talk: the setting, three questions, how I test them, five findings, and what is still running

The Problem

A limited, shared resource

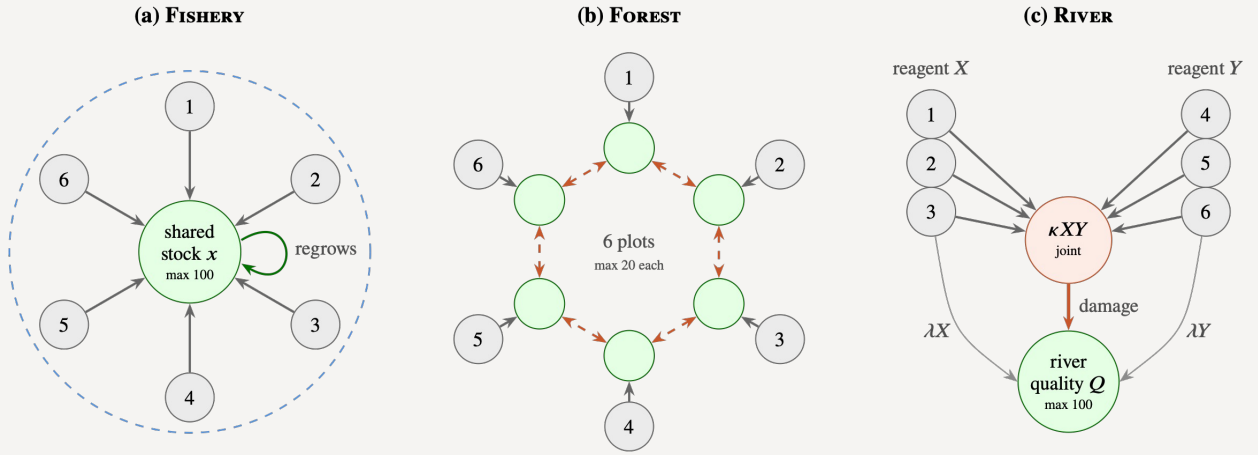
Each agent receives an allowance: the amount it may take this round. Taking more helps that agent and leaves less for everyone else.

Oversight by sampling

Checking every action costs too much, so an overseer spot-checks a few, as tax and fisheries inspectors do.

The open question

When does a sampled check stop agents taking more than their allowance, and how large does the penalty need to be?



The same question in three settings, chosen because one agent's over-use reaches others in a different way in each: through a single shared pool (Fishery), through damage to direct neighbours (Forest), and through harm that appears only when two kinds of use combine (River). A result that holds in all three does not rest on one structure.

Three Questions

01

What has to follow a catch?

When a check finds an agent taking more than its allowance, the overseer can fine it, remember it, or do nothing. Which of these reduces harm?

02

Can the stopping fine be predicted?

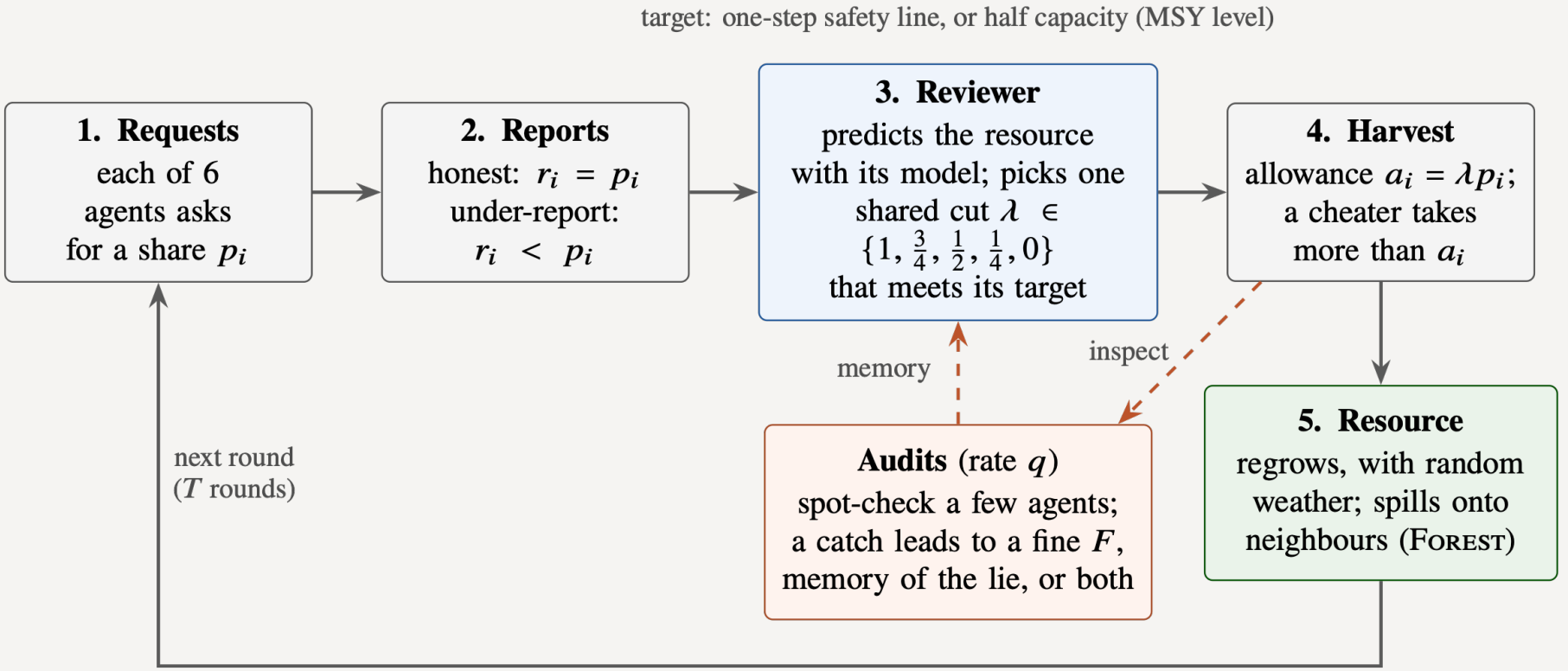
Is there a fine size at which over-taking stops, and can it be worked out before that fine is tried?

03

Do language models behave the same way?

When most agents are language models making their own choices, do they stop where the prediction says?

How One Round Works



Each round, every agent asks for a share; the reviewer predicts what the requests would do to the resource and scales them all by one factor; each agent then takes its allowance or more; the resource regrows; and a few agents are checked at random.

Harm = the resource falling below half its capacity, the level at which it regrows fastest

How I Test It

01 **Run the setting**

Six agents share the resource for many rounds; the reviewer sets allowances; a few agents are checked at random each round.

02 **Measure the gain**

Run games without any fine and record how much an agent gains from one round of taking more than its allowance.

03 **Predict, then freeze**

Work out the fine that should stop over-taking and record the prediction before any game with that fine is run.

04 **Test on games the prediction never saw**

Run the predicted fine and the fines around it on new games, and report the result whether or not it matches.

Key quantities

q	chance that an agent is checked in a round (1 in 6 for the language models)
F	fine paid when a check finds an agent over its allowance
$e = q \times F$	expected fine for one round of over-taking
g	what an agent gains from one round of over-taking
$e > g$	the prediction: over-taking stops once the expected fine is larger than the gain (Becker 1968)

Finding 1: A Catch Needs a Consequence

The setting

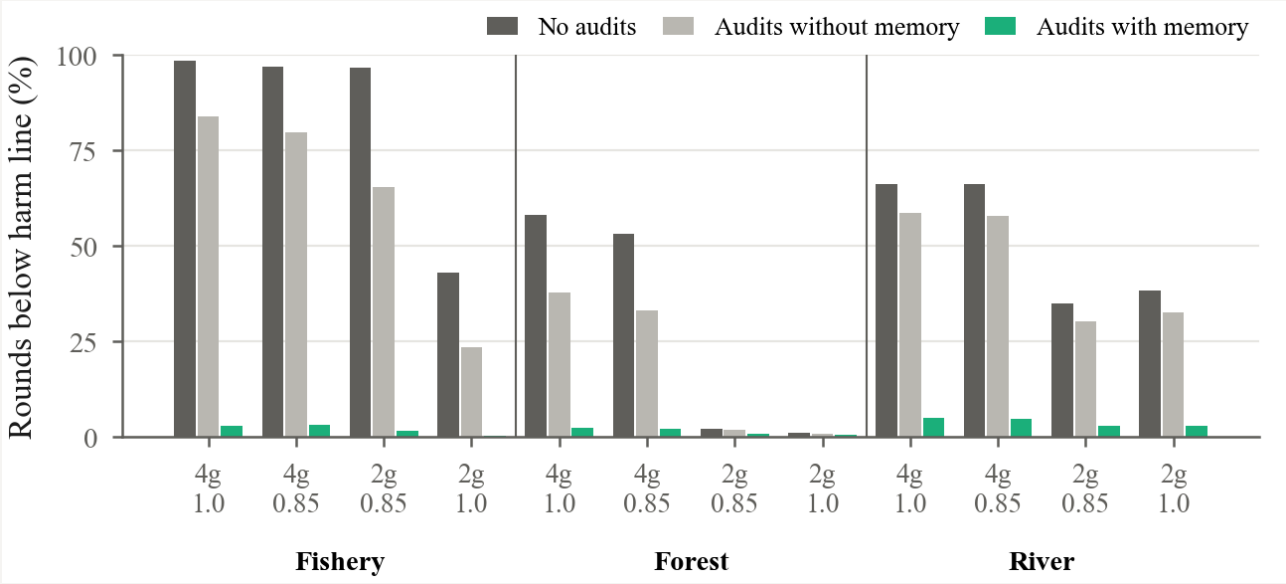
Some agents report less than they intend to take, so the reviewer underestimates their use unless a check catches them.

Checks without memory

Correcting only the round in which a lie was caught removed 10–46% of the harm.

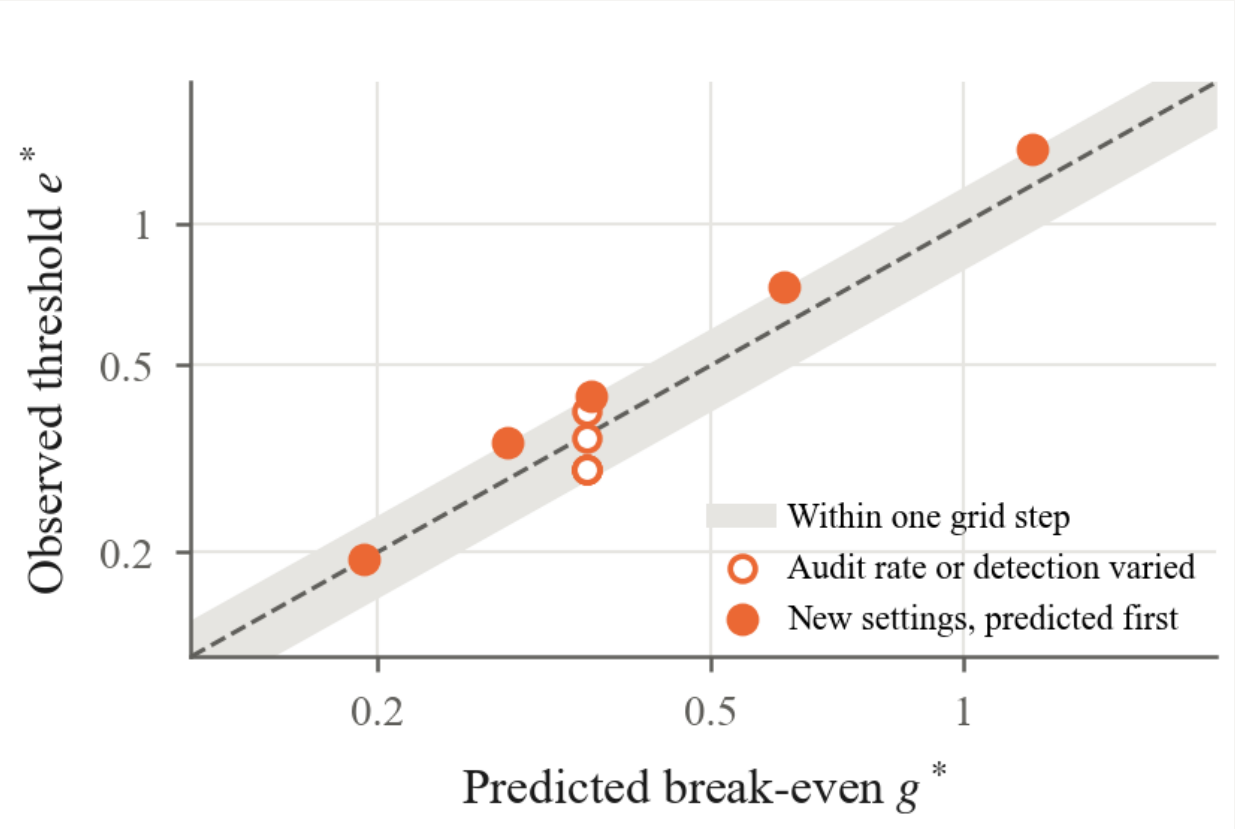
Checks with memory

Remembering each caught lie and adjusting that agent's later allowances reduced harm in all 12 settings tested. In Fishery, rounds below half capacity fell from 43–99% to 0.3–3%.



Share of rounds below half capacity when some agents under-report. Dark: no checks. Light: checks without memory. Green: checks with memory. Each group has four settings: 2 or 4 misbehaving agents (2g, 4g) and two regrowth rates (1.0 and 0.85 of normal).

Finding 2: A Predictable Threshold



Observed against predicted stopping point (expected fine). Filled: new settings predicted in advance. Open: the checking rate, or the chance that a check catches the cheat, was varied. The band marks one step of the tested grid.

What was predicted

Programmed agents choose how much to over-take so as to earn the most. From games without checks, we predicted the expected fine at which they would stop.

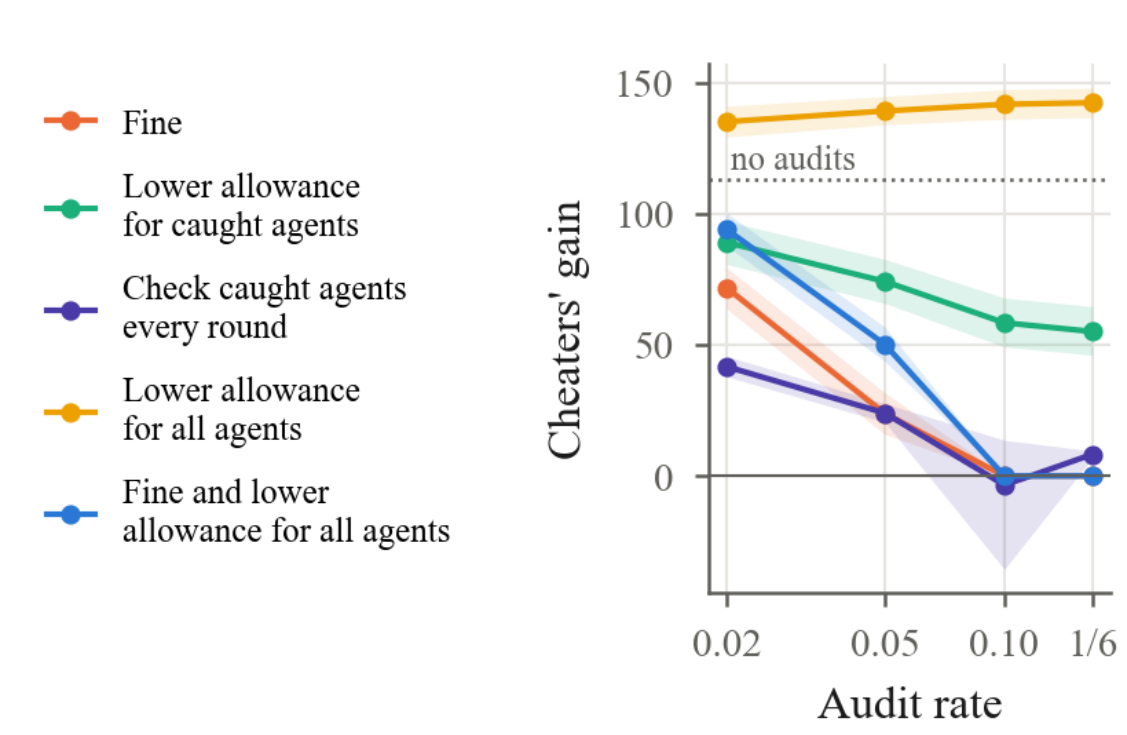
The result

In 5 of 5 new settings, the observed stopping point was within one step of the prediction.

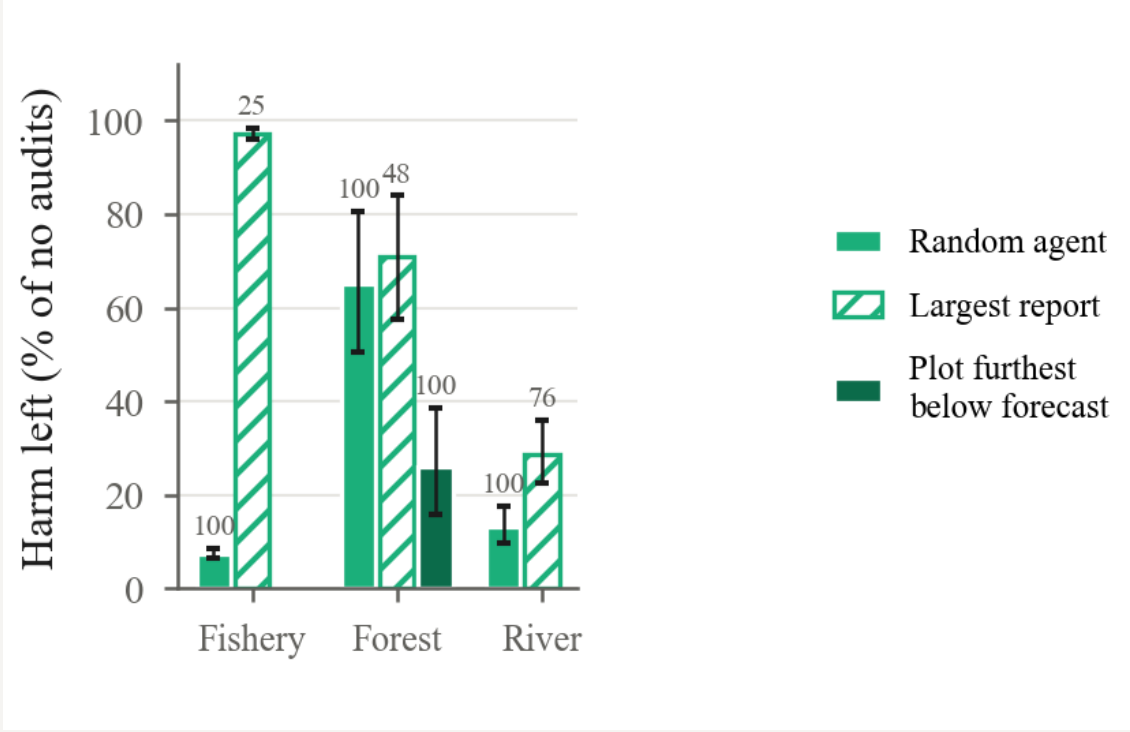
Why this comes first

These agents are built to maximise their payoff, so this checks the measurement method before it is used on language models.

Finding 3: Two Shortcuts That Backfire



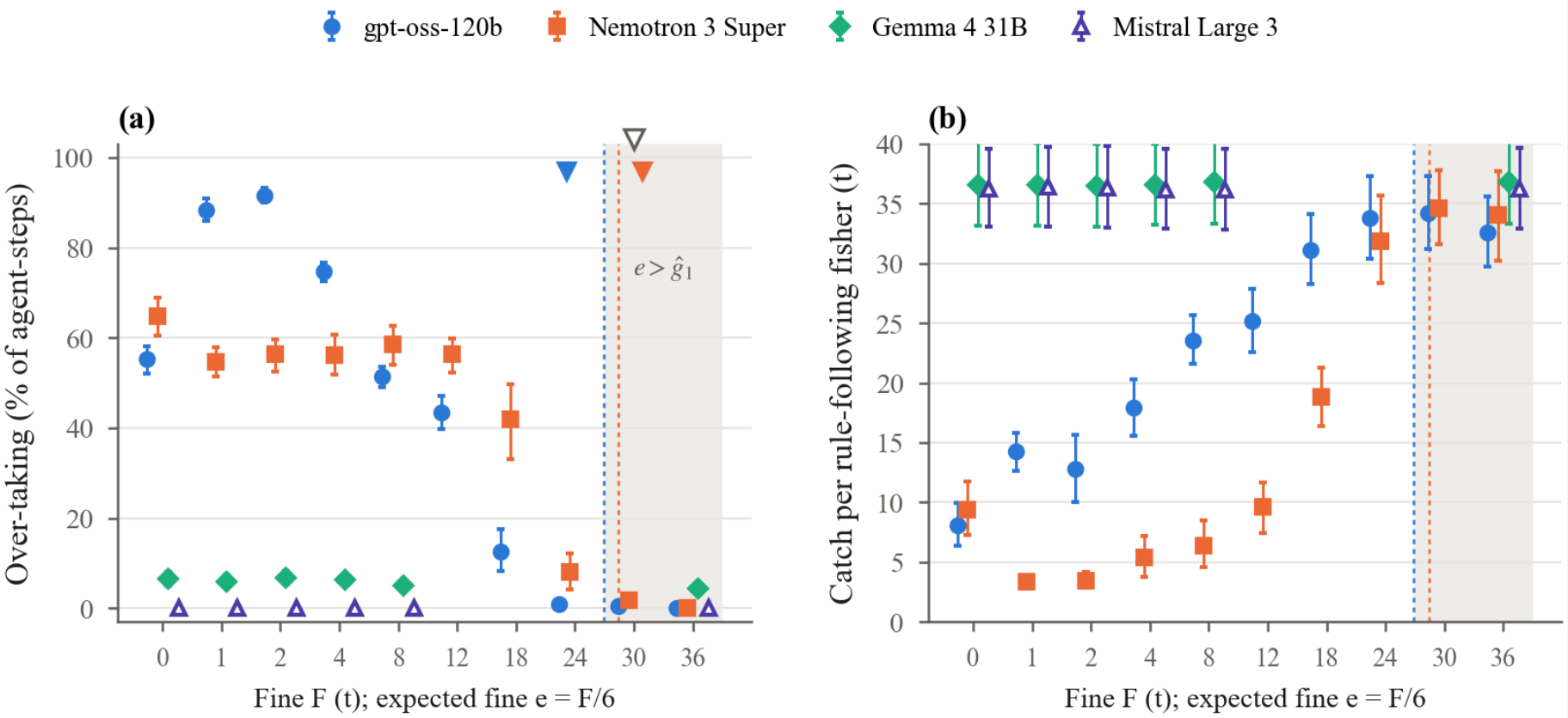
Lowering every agent's allowance to make up for known cheaters let the cheaters gain more than with no checks at all (142 vs 113); honest agents paid for it.



Always checking the agent with the largest report caught 25% of liars over a game, against 100% with random checks: the largest reporter is often honest.

Shortcuts that save checks can open blind spots

Finding 4: Language Models Stop Where Predicted



Four open language models each controlled four of the six fishers (520 games). Left: share of rounds in which a model took more than its allowance, at each fine, with a 1 in 6 chance of a check. Triangles: stopping fine predicted before testing (open) and observed (filled). Right: catch left for the two rule-following fishers.

Predicted before testing: 30 t. Observed: 30 t (Nemotron 3 Super), 24 t (gpt-oss-120b)

Finding 5: One Round's Gain Matters

What predicted it

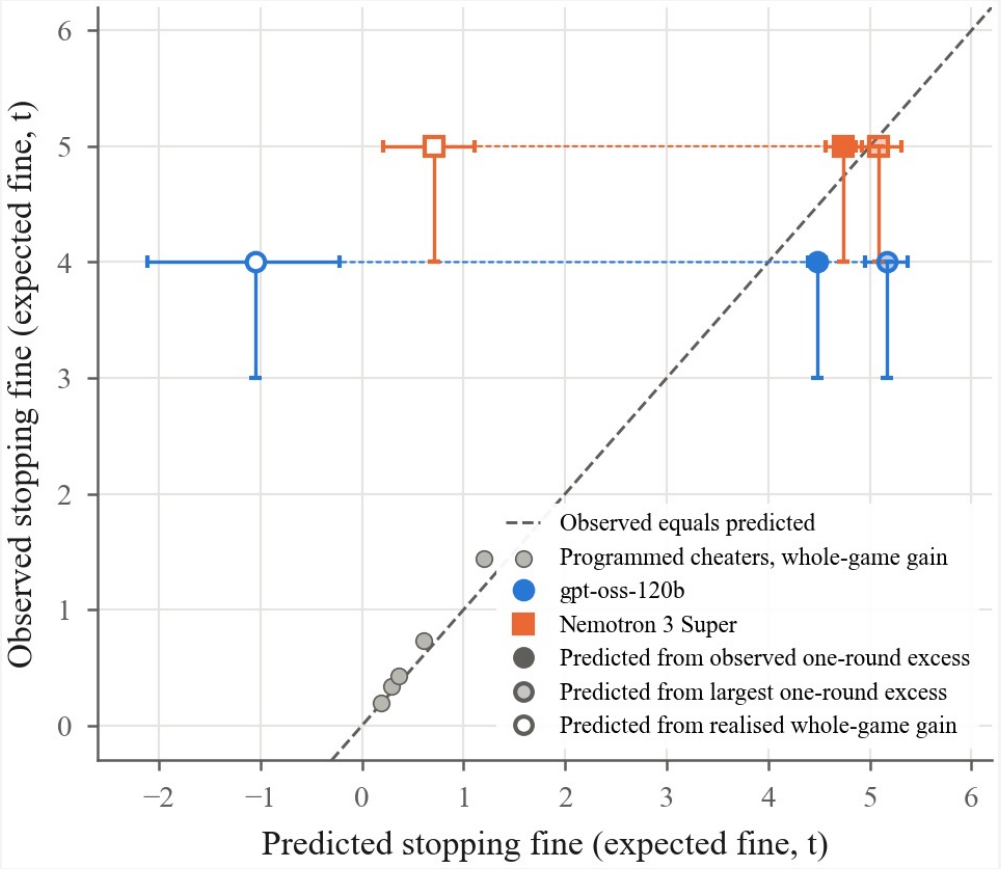
How much each model took above its allowance in one round without a fine (4.5–4.7 t).

What did not

What over-taking earned over a whole game: about −1 to +0.7 t per round, because over-taking depleted the lake.

Compared with an ideal optimiser

An agent that maximised its expected payoff would keep over-taking until 36 t. Both models stopped earlier, and gradually.



Each point compares a predicted and an observed stopping fine; points on the dashed line were predicted exactly.

Memory Works for Language Models Too

	Checks only		Checks + memory
Rounds with over-taking	55%	→	**20%**
Rounds below half capacity	92%	→	**20%**
Lakes collapsed (of 10)	5	→	**0**

gpt-oss-120b, no fine in either condition, 10 games each

Reduced, not stopped

Memory protected the lake without any fine, but some over-taking remained. For Nemotron 3 Super, over-taking fell from 65% to 33%.

With a large enough fine

Fines above the gain stopped over-taking and kept the harvest at 0.99 of the maximum the lake can sustain.

Running Now (Results Soon)

Expected fine, or the size of the fine?

Same expected fine (2 t or 3 t), made up differently: 1 in 3 checked with a small fine, 1 in 6, or 1 in 12 with a large fine. gpt-oss-120b, 20 new games per condition.

Result (to fill in):

Does the prediction hold on new games?

Stopping fine recorded in advance (30 t for both models), tested on 10 new games and with the rules reworded.

Result (to fill in):

What if nothing is ever checked?

Rules that never mention checks, compared with checks that carry no fine.

Result (to fill in):

Does over-taking spread to neighbours?

Forest: one programmed over-taker among five language-model agents; neighbours' cuts either hidden or visible.

Result (to fill in):

Where This Is Going

Sep – 8 Oct

Programmed agents in all three settings; language models in Fishery

12 – 14 Oct

Follow-up tests asked for by two outside reviews

20 Nov

Workshop submission (AAAI-27 WMAC) and arXiv

Jan 2027

Main conference: IJCAI or ICML

Each study has a protocol written down before its code; predictions that failed are reported.

Questions for You

01 Is "taking more than your allowance" a fair stand-in for an AI agent misbehaving? What would be better?

02 If a fine works under one wording of the rules, how far would you trust it under another?

03 What would convince you more: more models, or more kinds of setting?

04 Where in your own work does someone check only a sample, and what happens after a catch?

Thank You

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